Model Selection: Why the Two-Stage Hurdle Model

Poisson GLM

What it is:

A generalized linear model that treats crash counts as Poisson-distributed. It fits a linear combination of features through a log link: `log(λ) = β + βx + ... + βx`, then predicts `λ = e^(linear combination)`. The Poisson distribution has one parameter λ which serves as both the mean and the variance.

Why it failed:

- The Poisson assumption is `mean = variance`. In this dataset the variance of crash counts is 287× the mean — the data is massively overdispersed. The GLM's loss function is mis-specified from the start; it's penalizing predictions under a distribution that doesn't describe the data.
- With 98.47% zeros, the log-likelihood is dominated by zero windows. The model's optimal strategy under Poisson loss becomes "predict a very small λ everywhere." This minimizes total error but produces a flat, near-zero risk surface that is useless for ranking — every segment looks equally risky.
- It is linear in the features after the log transform. It cannot learn interactions like "freezing AND arterial AND school zone AND rush hour" without those being manually specified.

Negative Binomial GLM

What it is:

An extension of the Poisson GLM that adds a dispersion parameter `α`, allowing variance to exceed the mean: `variance = λ + α·λ²`. This relaxes the mean=variance constraint.

Why it failed:

- It handles overdispersion correctly but still models all zeros with a single mechanism. In this data there are two types of zero: *structural zeros* (segments where crashes almost never happen regardless of conditions) and *sampling zeros* (segments where crashes do happen but didn't during this specific window). The NB GLM can't distinguish these — it treats every zero as the same kind of absence.
- Because it doesn't separate occurrence from intensity, it still over-weights the sea of zeros during

fitting. The result is better calibrated on the full count distribution but still poor at ranking — it can't reliably identify the top 5% of high-risk windows.

- Still linear in features after the log transform. Same interaction limitation as Poisson GLM.

Zero-Inflated Negative Binomial (ZINB)

What it is:

A two-component mixture model that explicitly separates zeros into two sources. Component 1 is a binary model that generates "always-zero" outcomes (structural zeros). Component 2 is a Negative Binomial that generates counts including zero (sampling zeros). The two components are fit jointly via EM or maximum likelihood.

Why it failed:

- Conceptually the right idea — it acknowledges the two-process structure. But the parameters of both components are estimated simultaneously under a shared likelihood. This joint optimization is sensitive to initialization and tends to have flat or multimodal likelihood surfaces, making convergence unreliable.
- No native gradient boosting implementation exists. ZINB is a parametric GLM — it fits linear predictors, so it shares the same feature interaction limitation as the Poisson and NB GLMs.
- Slower to train and harder to tune than tree-based models. With 66 features and hundreds of thousands of rows, the EM fitting becomes computationally expensive and unstable.
- The end result is similar to hurdle in structure but worse in practice because it lacks the predictive power of gradient boosting and requires careful statistical tuning for each dataset.

XGBoost (count regression)

What it is:

A gradient boosted tree ensemble that optimizes a differentiable loss over sequential shallow trees. Far more flexible than any GLM — it can learn arbitrary non-linear interactions between features without manual specification. Can be configured with a Poisson or squared error loss to predict counts.

Why it failed:

- The core problem: it uses a single model to predict the count directly, including all zeros. At 98.47% zeros, gradient updates are dominated by the majority class. The trees learn mostly "when is this a zero?" and get very little gradient signal from the rare high-count events that actually matter for routing.
- It doesn't natively separate occurrence from intensity. To predict a window with 5 crashes, the same trees must also correctly predict the 98.47% of windows with 0 crashes. These two tasks have partially

conflicting gradient signals — features that push predictions toward 0 for non-crash windows also suppress predictions for crash windows.

- Unlike the hurdle, there is no Stage 2 that trains exclusively on crash windows. The rare 5-crash events contribute ~1.5% of the gradient. They are statistically invisible to the optimization.

Why the Hurdle Wins

Each rejected model fails for one of two reasons: either it **misspecifies the distribution** (Poisson, NB) or it **conflates two separate problems** (ZINB without boosting, XGBoost single-stage). The hurdle solves both:

- Stage 1 uses gradient boosting purely for the binary question, with the full 98.47% zero signal directed at one task — no interference from count estimation
- Stage 2 uses gradient boosting purely for the count question, trained only on crash windows — the distribution it sees is no longer zero-inflated
- The tree ensemble in both stages captures non-linear feature interactions that no GLM can
- Isotonic calibration is applied post-hoc on a held-out set — something GLMs don't need but that tree classifiers specifically require because their outputs are scores, not probabilities